OPERATING SYSTEMS

UE25CS242A

PES UNIVERSITY

CELEBRATING 50 YEARS

File Management

Pavan A C

Department of Computer Science & Engineering

OPERATING SYSTEMS

OS Security Problems: Program Threats

Pavan A C
Department of Computer Science & Engineering

OPERATING SYSTEMS

Slides Credits for all PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

The Security Problem



Why is Security Important?

- An Operating System manages valuable resources such as:
  - Programs and processes
  - Files and data
  - Memory and CPU
  - Network resources
  - User accounts and credentials
- A security breach can result in:
  - Unauthorized access
  - Data theft or disclosure
  - Data modification or destruction
  - Misuse of system resources
  - Disruption of services

The Security Problem



Types of Security Violations

1. **Breach of Confidentiality -** Unauthorized reading or disclosure of data.
2. **Breach of Integrity -** Unauthorized modification of data.
3. **Breach of Availability -** Unauthorized destruction or disruption of resources.
4. **Theft of Service -** Unauthorized use of system resources.
5. **Denial of Service (DoS) -** reventing legitimate users from accessing a system or service.

OPERATING SYSTEMS

Program Threats



- Program threats occur when a malicious or compromised program attempts to gain unauthorized access, misuse resources, or damage a computer system.
- Programs and processes are the primary means through which work is performed on a computer.
- An attacker can exploit a program to:
  - Access unauthorized data
  - Modify or delete files
  - Gain additional privileges
  - Execute malicious code
  - Disrupt system operation

Common Program Threats - Malware, Trojan Horse, Spyware, Ransomware, Code, Injection, Viruses, Worms, Trap Doors / Back Doors

OPERATING SYSTEMS

Malware and Trojan Horse



Malware - Malware (Malicious Software) is software designed to exploit, disable, damage, or gain unauthorized access to a computer system.

- Malware may perform malicious actions while appearing to be a legitimate program.
- Common forms include: Trojan Horse, Spyware, Ransomware, Viruses, Worms

Trojan Horse

- A Trojan Horse is a program that appears to perform a legitimate function but secretly performs malicious activities.
- It may run with the user's privileges, allowing it to access resources available to that user.
- The user may unknowingly install or execute the program.

Spyware



Spyware

- Spyware is malicious software that secretly collects information about a user or system.
- It may monitor:
  - Browsing activity
  - User behavior
  - Credentials and personal information
  - Files and system information
- Collected information may be sent to a remote attacker/server without the user's knowledge.

Example: A malicious application secretly records user activity and sends the collected information to an attacker.



Ransomware

- Ransomware is malware that encrypts files or data on a system.
- The attacker demands a ransom payment in exchange for the decryption key.
- Without the key, the victim may be unable to access the encrypted data.

OPERATING SYSTEMS

Code Injection



- Code injection is a security attack in which an attacker causes unauthorized code to be inserted into or executed by a program.
- It commonly occurs due to poor or insecure programming practices.
- If successful, the attacker may:
  a. Modify program behavior
  b. Execute unauthorized commands
  c. Access protected data
  d. Gain control over the program

Common Example: Buffer Overflow

- A program stores data in a fixed-size buffer.
- If more data is supplied than the buffer can hold, the extra data may overwrite nearby memory.
- Carefully crafted input can potentially alter program execution.

Viruses



- A virus is a fragment of code embedded in a legitimate program or file.
- It is designed to replicate itself by infecting other programs or files.
- A virus generally requires user action or execution of an infected program to spread.
- Viruses can:
  a. Modify or destroy files
  b. Corrupt programs
  c. Cause system crashes or malfunction
  d. Spread to other files or systems

Worms



- A worm is a self-replicating program that can spread across computer systems, typically through a network.
- Unlike a virus, a worm does not need to attach itself to another program or file.
- Worms can spread automatically by exploiting network or software vulnerabilities.
- They may consume system and network resources and cause:
  a. Network congestion
  b. System slowdown
  c. Service disruption
  d. Installation of additional malware

OPERATING SYSTEMS

Trap Door / Back Door & Principle of Least Privilege



- A trap door (back door) is a hidden entry point in a program that allows access to a system without following the normal authentication or security mechanisms.
- It may be intentionally created during development or maliciously inserted by an attacker.
- If discovered and exploited, it can provide unauthorized access to system resources.

Principle of Least Privilege

- A user or program should be given only the minimum privileges required to perform its task.
- Privileges should be limited in:
  - Scope — what resources can be accessed
  - Time — how long access is required



THANK YOU

Pavan A C

Department of Computer Science and Engineering

pavanac@pes.edu